llmXive Automated Review of RynnWorld-Teleop: An Action-Conditioned World Model for Digital Teleoperation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a novel "digital teleoperation" paradigm using an action-conditioned world model. The central claims regarding the system's ability to generate high-fidelity videos and train policies are generally supported by the provided tables and figures. However, there are specific inconsistencies between the mathematical definitions and the textual descriptions, as well as minor ambiguities in the reporting of performance metrics that need resolution.

First, in Section 3.1 (Preliminaries), the text states that the network v_Θ is trained to predict the "velocity field" following the Conditional Flow Matching (CFM) framework. Equation 1 defines the probability path $z_t = (1-t)z_0 + t\epsilon$, which implies the velocity is $\epsilon - z_0$. However, Equation 2 defines the loss target as $(\epsilon - z_0)$. In standard CFM, the velocity is indeed $\epsilon - z_0$, but the text's phrasing "predict the velocity field" combined with the equation $v_\Theta(z_t, t, z_{ref}, c) - (\epsilon - z_0)$ suggests a potential confusion in terminology or a missing scaling factor of $1/t$ if the target was intended to be the instantaneous velocity vector field at time t. The authors should clarify if the target is the constant velocity vector or the instantaneous velocity, as this affects the interpretation of the training objective.

Second, regarding the performance claims in Section 4.2 and Table 3, the text highlights a throughput of "40.0 fps" for the distilled causal student. While Table 3 confirms this number, the text describes the inference as using a "4-step flow matching schedule." It is crucial to explicitly state that the 40 fps measurement includes the full 4-step generation process. If the 40 fps were measured per-step or under different conditions, the claim of "real-time interactive generation" (which typically requires >30 fps end-to-end) might be misleading. The breakdown of latency (5% encoding, 72% denoising, 23% decoding) sums to 100% of the 25ms frame time, but the text should explicitly confirm that the 40 fps figure corresponds to the 4-step sampling process described.

Finally, the claim in Section 4.2 that the system "significantly exceeds the typical 2–10 Hz frame rates of existing action-conditioned world models" is supported by the comparison in Table 3 (where baselines like InterDyn and CosHand are ~2.9 fps and 0.8 fps). However, the text cites "Wang et al. 2026" and "Akkerman et al. 2025" for these baselines. While the table lists the FPS, the text should ensure the comparison is apples-to-apples (e.g., same resolution, same hardware) or clarify if the speedup is due to the distillation method specifically. The current presentation is strong but would benefit from a brief note on the experimental conditions for the FPS comparison to fully substantiate the "significant exceedance" claim.

These issues are primarily clarifications of existing data rather than fundamental flaws in the science, warranting a minor revision.

Action items.

- **[writing]** The paper presents a novel "digital teleoperation" paradigm using an action-conditioned world model. The central claims regarding the system's ability to generate high-fidelity videos and train policies are generally supported by the provided tables and figures. However, there are specific inconsistencies between the mathematical definitions and the textual descriptions, as well as minor ambiguities in the reporting of performance metrics that need resolution. First, in Section 3.1 (Preliminarie

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively contrasts physical and digital teleoperation workflows, but the diagram labels do not explicitly match the 'world model' terminology used in the caption, and the caption text contains a formatting error.

Figure 2

The figure fails to visually demonstrate the 'Depth-Aware Representation' described in the caption; instead of clear skeletal renderings with depth modulation, it shows indistinct, low-resolution point clouds that do not support the claim.

Figure 3

The figure provides a clear visual overview of the pipeline architecture, but the caption contains missing model names (placeholders), and the input illustration does not match the described 'skeletal video' input.

Figure 4

Figure 4 effectively illustrates the four manipulation tasks (Dual Picking, Block Pushing, Bimanual Lifting, Lid Placement) designed for evaluation. The sub-panels are clearly labeled, and the red arrows and numbered points provide intuitive visual cues for the specific actions and interaction points without requiring a complex legend.

Figure 5

The figure effectively visualizes the overlap between real and synthetic data distributions, but the caption contains a missing model name and the plots lack axis labels and scales.

Figure 6

The figure displays a grid of robotic manipulation videos, but the caption is marred by missing model names and makes a claim about 'human and robotic' embodiments that is not visually supported by the provided image.

Figure 7

The figure effectively demonstrates visual generalization to new objects and backgrounds, but the caption is grammatically incomplete due to missing model names and lacks specific row-by-row descriptions for the four displayed sequences.

Figure 8

The figure fails to provide the quantitative evidence necessary to support the caption's claims regarding stability and performance differences between conditioning strategies, appearing instead as a qualitative visual comparison without metrics.

Figure 9

Figure 9 effectively illustrates the hardware infrastructure by displaying the TIANJI M6 robot and WUJI Hand alongside clear text boxes detailing their specific technical specifications (DOF, payload, weight, force). The visual presentation is uncluttered, legible, and directly supports the caption's claim of summarizing detailed specifications for reproducibility.

Figure 10

The figure is a low-quality, cropped photograph that fails to provide a clear or verifiable view of the operator setup, with critical visual information obscured by a smiley face and a lack of internal labels.

Figure 11

The figure presents a grid of qualitative results but fails to visually demonstrate the specific comparison claimed in the caption regarding mask-based versus skeletal conditioning. Additionally, the layout contains unexplained white gaps in the baseline rows that contradict the caption's note about native resolutions.

Figure 12

The figure provides a clear visual demonstration of the model's video synthesis capabilities across four tasks, but the caption is grammatically broken with missing model names, and the image lacks labels to identify the specific tasks shown in each row.

Action items.

- **[science]** Figure 1: The caption states that digital teleoperation replaces the real robot with 'a real-time action-conditioned world model', but the diagram labels the central component as 'Real-time Video Streaming' and the box as 'Synthetic Dataset', failing to explicitly name or visualize the 'world model' described in the text.

- **[writing]** Figure 1: The caption contains a formatting error 'operator-hours $$ hardware availability' where a mathematical symbol or word (likely 'and' or 'plus') is missing or corrupted.
- **[science]** Figure 2: The caption claims 'depth-modulated color and size' for hand skeletons, but the rendered image displays only static, low-resolution 3D point clouds (blue/red blobs) without visible skeletal structures or clear depth-based modulation.
- **[writing]** Figure 2: The image is extremely low-resolution and pixelated, making it impossible to verify the specific rendering details (skeletons, size modulation) described in the caption.
- **[writing]** Figure 3: The caption contains multiple instances of missing model names (e.g., 'Overview of .', 'replaces the real robot with ,'), likely due to a placeholder variable not being rendered in the text.
- **[science]** Figure 3: The diagram shows a 'VAE Encoder' processing 'depth-aware skeletal videos' (bottom left), but the input visual is a cartoon character and whiteboards, which does not match the caption's description of 'hand-pose conditioning' or 'skeletal videos'.
- **[writing]** Figure 3: The label 'Real-time' in the bottom right output panel is extremely small and illegible, making it difficult to read without zooming in significantly.
- **[writing]** Figure 5: The caption contains a missing model name (indicated by a blank space before '-generated trajectories'), making it unclear which system generated the synthetic data.
- **[science]** Figure 5: The t-SNE plots lack axis labels and numerical scales, preventing assessment of the embedding space dimensions or cluster separation metrics.
- **[writing]** Figure 6: The caption contains multiple instances of missing model names (e.g., 'Qualitative Results of .', 'absorbs rich manipulation priors'), likely due to a placeholder variable not being replaced with the actual model name.
- **[writing]** Figure 6: The caption claims the model maps gestures to 'both human and robotic embodiments,' but the figure only displays robotic execution videos; a human embodiment comparison is missing.
- **[writing]** Figure 7: The caption contains multiple instances of missing model names (e.g., 'generalizes along two axes', 'absorbs rich manipulation priors', 'preserves high visual fidelity') where the subject should be explicitly stated.
- **[science]** Figure 7: The figure displays four rows of video frames, but the caption only describes two axes of generalization (unseen objects in top two rows, unseen backgrounds in bottom two rows) without clarifying the specific distinction between the two rows within each category.

- **[science]** Figure 8: The figure displays two images labeled 'Concat' and 'Add' but lacks the quantitative data (e.g., loss curves, fidelity metrics, or error bars) required to support the caption's claim that the additive scheme ensures 'stable synthesis' compared to concatenation.
- **[writing]** Figure 8: The caption references 'Eq. [missing]' for the additive scheme, but the equation number is missing from the text.
- **[science]** Figure 10: The image is a low-resolution, heavily cropped photograph with a smiley face obscuring the operator's head, making it impossible to verify the specific hardware or setup details claimed in the caption.
- **[writing]** Figure 10: The image contains no internal labels, scale bars, or annotations to identify the 'operator setup' components, relying entirely on the generic caption for context.
- **[writing]** Figure 11: The caption states 'Unlike baselines that rely on SAM-generated masks... uses sparse skeletal poses,' but the figure displays 'Control Mask' (masks) and '2D skeletal' (poses) as separate rows rather than showing the baselines using masks as conditioning inputs. The visual evidence does not support the specific comparison described in the text.
- **[writing]** Figure 11: The caption claims 'baselines are visualized at their native supported resolutions,' yet the 'CosHand' and 'Mask2IV' rows contain large white vertical gaps between frames, which appears to be a layout artifact or missing data rather than a native resolution characteristic.
- **[writing]** Figure 12: The caption contains multiple grammatical errors and missing nouns (e.g., 'Qualitative results of on robotic...', 'synthesizes high-fidelity...'), likely due to a missing model name placeholder.
- **[writing]** Figure 12: The image consists of four rows of video frames but lacks any row labels or task descriptions to identify the specific manipulation tasks being demonstrated.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written and defines most specialized terms (e.g., "digital teleoperation," "depth-aware action representation") clearly upon introduction. However, there are a few instances where notation and acronyms are introduced without sufficient context for a competent reader from an adjacent field (e.g., a computer vision researcher not deeply embedded in the specific diffusion distillation subfield).

Specifically, the symbol `z_V` in Section 3.1 appears abruptly in an equation without a preceding definition, creating a momentary stall for the reader trying to map the notation to the text. Additionally, while "CFM" and "DMD" are standard in the immediate subfield of generative modeling, their specific variants ("causal flow-matching") and the term "sink token" could benefit from brief operational definitions to ensure the reader understands the *mechanism* implied by the jargon, not just the name. The use of "SFT" without expansion in Section 4.1 is a minor oversight compared to the rest of the paper's rigor.

Addressing these specific points—defining `z_V`, clarifying the relationship between CFM and causal flow-matching, and expanding SFT—will ensure the paper is fully self-contained for the target audience of a competent adjacent-field PhD.

Action items.

- **[writing]** Section 3.1 (Preliminaries): The symbol `z_V` appears in the equation `z_0 = \mathcal{E}(I_{V})` without definition. It is unclear if `I_V` refers to a specific video frame, a variable, or a typo for `I_{ref}`. Define `I_V` or correct the notation to match the reference image `I_{ref}` used elsewhere.
- **[writing]** Section 3.1 (Preliminaries): The term 'conditional flow matching (CFM)' is introduced with an acronym, but the paper later uses 'causal flow-matching' (Section 3.4) without clarifying if this is a specific variant of CFM or a distinct objective. A brief gloss distinguishing the two or confirming they are the same framework would prevent confusion for adjacent-field readers.
- **[writing]** Section 3.4 (Autoregressive Distillation): The term 'Distribution Matching Distillation (DMD)' is used with an acronym, but the specific mechanism (e.g., score-based gradient guidance) is described only after the acronym is introduced. Define DMD at first use as 'Distribution Matching Distillation (DMD), a technique that...' to ensure the reader understands the method before encountering the acronym.
- **[writing]** Section 3.4 (Autoregressive Distillation): The phrase 'persisted sink token' is used to describe the embedding of `I_{ref}`. While 'sink token' is a known concept in some transformer literature, it is not universally standard. Add a brief parenthetical explanation (e.g., 'a persistent token that anchors the context') to ensure clarity for readers from adjacent NLP or vision fields.
- **[writing]** Section 4.1 (Training Details): The acronym 'SFT' (Supervised Fine-Tuning) is used without being explicitly defined at first use in the main text (it appears in the bullet points). While common in ML, explicitly spelling it out as 'Supervised Fine-Tuning (SFT)' at its first occurrence in Section 4.1 would align with the paper's otherwise rigorous definition style.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is logically sound and internally consistent. The central claim—that a generative world model can serve as a "digital teleoperation" system to decouple data collection from physical hardware—is supported by a coherent chain of reasoning: (1) the definition of the paradigm requires robot-centric, action-grounded, and real-time capabilities; (2) the proposed method (RynnWorld-Teleop) explicitly addresses these three requirements via depth-aware conditioning, progressive training, and autoregressive distillation; and (3) the experimental results (Sim2Real transfer and data augmentation) directly validate the utility of the generated data for policy learning.

There are no contradictions between sections. The dataset statistics in Table 1 (1,800 episodes) align with the description in Section 3.4 and the training details in Section 4.1. The distinction between the "EgoDex-Test" (human-centric) and "Robotic-Test" (robot-specific) benchmarks in Section 4.3 is clearly maintained throughout the results and ablation studies. The causal claims regarding the distillation process (Section 3.5) are supported by the ablation results in Table 3, which show performance drops when specific components (e.g., Causal Warm-up, DMD) are removed. The transition from the method description to the system evaluation is seamless, with the "chunked re-anchoring" strategy in Section 3.6 logically addressing the potential drift issue inherent in autoregressive generation. No non-sequiturs or scope inflation were detected; conclusions are appropriately qualified by the experimental scope (e.g., "across four distinct tasks").

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several broad claims regarding the scope and generality of its "digital teleoperation" paradigm that exceed the specific experimental evidence provided.

First, the abstract and introduction characterize the system as "embodiment-agnostic" and capable of transferring to "any target robot." However, the methodology (Section 3.4) explicitly relies on a "Robotic Domain Adaptation" stage where the model is fine-tuned on paired human-robot data specific to the target embodiment. The experiments (Section 4.1) are conducted exclusively on a single robot platform (TIANJI M6 with WUJI hands). There is no evidence presented of zero-shot transfer to a different robot family or kinematic structure. The claim of agnosticism is therefore unsupported; the method is "embodiment-adaptable" but requires specific fine-tuning data for each new robot, a significant constraint omitted from the high-level claims.

Second, the abstract states that policies trained "exclusively" on generated data achieve "effective zero-shot Sim2Real transfer." Table 4 reveals that while the method is competitive on some tasks (e.g., Block Pushing at 82.86% vs. 85.71% baseline), it underperforms significantly on others (Lid Placement at 28.57% vs. 57.14% baseline). Describing this mixed performance as "effective" without qualification overstates the results. The claim should be tempered to reflect that the method provides a "viable" or "partial" substitute, particularly for tasks with simpler dynamics, rather than a general solution.

Finally, the paper claims "real-time interactive generation" (Abstract, Introduction). While the inference speed is indeed 40 FPS (Section 4.2), the "Digital Teleoperation System" described in Section 3.5 utilizes "chunked re-anchoring" that requires the "actual egocentric frame from the robot's camera" to reset the reference image. This implies the system currently depends on a physical robot's camera feed to maintain long-horizon consistency, contradicting the narrative of a purely digital, hardware-independent loop where an operator teleoperates a virtual robot in real-time. The "real-time" claim conflates the model's inference speed with the system's operational requirements, which still necessitate physical hardware feedback for stability.

These issues are primarily rhetorical overreach regarding the scope of generalization and the nature of the "real-time" loop. They can be resolved by narrowing the claims in the abstract and introduction to match the specific conditions (single robot, fine-tuning required, mixed task performance) demonstrated in the experiments.

Action items.

- [**writing**] Abstract claims 'effective zero-shot Sim2Real' for policies trained exclusively on generated data. Table 4 shows this fails on 2/4 tasks (e.g., 28.57% vs 57.14% baseline on Lid Placement). Replace 'effective' with 'partial' or 'competitive on simple tasks' to match the mixed evidence.
- [**writing**] Abstract and Intro claim the method is 'embodiment-agnostic' and works on 'any target robot.' Experiments (Sec 4.1) use only one robot (TIANJI M6), and Sec 3.4 requires per-robot fine-tuning. Change 'embodiment-agnostic' to 'embodiment-adaptable' and clarify that transfer requires target-specific fine-tuning data.
- [**writing**] Abstract claims 'real-time interactive generation' for digital teleoperation. While inference is 40 FPS (Sec 4.2), the pipeline (Sec 3.5) requires 'actual egocentric frame from the robot's camera' for re-anchoring, implying physical hardware dependency. Clarify that 'real-time' applies to inference, not the full loop which currently needs physical feedback.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a method for "digital teleoperation" using a generative world model to synthesize robotic execution videos from human hand-pose streams. From a safety and ethics perspective, the work is low-risk. The system generates synthetic visual data for training imitation learning policies; it does not create operational exploits, biological hazards, or systems designed for deception or surveillance.

The data collection process involves human operators wearing motion capture trackers and data gloves (Sec. 3.2, Appendix). The paper describes the hardware setup and the retargeting pipeline but does not explicitly state IRB approval or informed consent procedures. However, the data consists of kinematic pose streams and synchronized video of the operator's hands in a controlled lab environment, not sensitive personal information or private conversations. In the context of robotics research where operators are typically lab members or paid participants in a controlled setting, the absence of a specific IRB statement is a minor disclosure gap rather than a fatal ethical violation, provided the data is not released in a way that re-identifies individuals. The paper does not release raw video of the operators, only the synthesized robot videos and the retargeted action vectors, which mitigates privacy risks.

There is no evidence of dual-use capabilities that lower the barrier to harm (e.g., the model does not generate instructions for weaponizing robots or bypassing safety protocols). The "digital teleoperation" paradigm is a data augmentation technique, not a remote control system for malicious actors. The paper appropriately focuses on the technical challenges of embodiment gaps and real-time generation. No specific safety disclosures or mitigations are required beyond standard research practices for human-subjects data in robotics, which are implicitly satisfied by the controlled experimental setup described.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling paradigm shift with "digital teleoperation," but the experimental design in the policy learning and baseline comparison sections lacks the rigor required to definitively support the headline claims of robustness and superiority.

First, the primary evidence for the system's utility—policy performance improvements in Table 2—is based on a single run of 35 trials per task. The reported gains (e.g., +5.7% for Dual Picking, +20% for Lid Placement) are presented as deterministic facts. However, in robotic policy learning, success rates on small test sets (n=35) exhibit high variance depending on the random seed used for policy initialization and data shuffling. Without reporting standard deviations or results across multiple independent training seeds, it is impossible to distinguish a genuine methodological improvement from a lucky initialization or a favorable random seed. The claim that the method "consistently improves" success rates is not statistically supported by the current single-point estimates.

Second, the quantitative comparison of the world model itself (Table 1) suffers from an unfair baseline configuration. The proposed method is evaluated at 480x832 resolution with a 4-step sampling schedule, while the baselines (InterDyn, CosHand, Mask2IV) are evaluated at significantly lower resolutions (256x256 or 320x512) and with 50 DDIM steps. Since video quality metrics like PSNR and SSIM are heavily dependent on resolution and sampling steps, the reported superiority of RynnWorld-Teleop is confounded by these hyperparameter differences. The design fails to isolate the contribution of the proposed depth-aware conditioning from the simple advantage of higher resolution and more inference steps. To validate the architectural contribution, baselines must be re-evaluated under identical resolution and step constraints.

Finally, the "Zero-Real-Data" claim for π_0 (Table 2) is particularly sensitive to variance. Achieving 68-82% success on complex tasks with *only* synthetic data is a strong claim that requires robust statistical backing. The absence of error bars or multiple seeds makes this result appear fragile. A skeptical reader cannot rule out that this performance is an outlier.

To close these gaps, the authors should: (1) re-run policy training with at least 3 different random seeds and report mean ± SD for all success rates; (2) re-evaluate all baseline world models at the same resolution (480x832) and step count (4 steps) to ensure a fair comparison; and (3) explicitly state the variance in the text or tables to demonstrate that the observed gains are statistically significant.

Action items.

- [**writing**] The paper presents a compelling paradigm shift with "digital teleoperation," but the experimental design in the policy learning and baseline comparison sections lacks the rigor required to definitively support the headline claims of robustness and superiority. First, the primary evidence for the system's utility—policy performance improvements in Table 2—is based on a single run of 35 trials per task. The reported gains (e.g., +5.7% for Dual Picking, +20% for Lid Placement) are presented as dete

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the experimental section requires minor corrections to align with the precision of the underlying data and to properly quantify uncertainty.

First, **Table 1** (Real-world policy performance) reports success rates with two decimal places (e.g., 88.57%). These results are derived from 35 trials per task. The standard error for a proportion $p \approx 0.88$ with $N = 35$ is $\sqrt{p(1-p)/N} \approx 0.045$ (4.5%). Reporting values to 0.01% precision (e.g., 88.57%) implies a false precision of three orders of magnitude beyond the actual statistical resolution. The authors should round these to integers or report as $88.6 \pm 4.5\%$.

Second, the text in Section 4.2 claims "consistent performance gains" and highlights specific improvements (e.g., a 20% increase in Lid Placement success rate) but provides no measure of uncertainty (standard deviation or standard error) or statistical significance testing. With $N = 35$, a 20% absolute difference is likely significant, but without a test (such as McNemar's test for paired binary outcomes or a bootstrap confidence interval), the claim of "consistent" improvement is anecdotal. The authors should either report the standard error of the mean for each condition or perform a significance test to validate the "consistent" claim.

Third, **Table 2** (Quantitative results of RynnWorld) reports video generation metrics (PSNR, SSIM, LPIPS, FVD) to 2-3 decimal places. In generative modeling, these metrics are known to have non-negligible variance across different random seeds or data splits. Reporting single-point estimates without standard deviations or confidence intervals prevents the reader from assessing the stability of the model's performance. The authors should report these metrics as mean ± standard deviation over at least 3-5 random seeds.

These are reporting issues that can be fixed by re-calculating summary statistics from the existing experimental runs; no new experiments are required.

Action items.

- [**writing**] Table 1 (tab:sim2real) reports success rates to two decimal places (e.g., 88.57%) based on N=35 trials per task. This implies a precision of ~0.03%, which is statistically unjustified given the binomial variance (SE 4.5% for p=0.88). Report results as integers or with ±1 standard error, and remove false precision.
- [**writing**] Section 4.2 claims 'consistent performance gains' and highlights specific improvements (e.g., +20% for Lid Placement) without reporting uncertainty metrics (SD/SE) or statistical significance tests (e.g., McNemar's test for paired proportions). Add error bars or p-values to support the claim of 'consistent' improvement.
- [**writing**] Table 2 (tab:exp) reports FVD, PSNR, and SSIM to 2-3 decimal places for video generation metrics. These metrics typically exhibit high variance across seeds. Report mean ± standard deviation over multiple random seeds (e.g., 3-5) rather than single-point estimates to allow assessment of stability.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the narrative flow is strong, particularly in the Introduction where the problem and solution are clearly delineated. However, there are several instances where sentence construction, redundancy, or undefined variables create minor friction for the reader.

In Section 3.1, the variable I_V appears in an equation without prior definition, while the surrounding text refers to I_{ref}. This forces the reader to pause and infer that I_V is the reference image. Consistency in notation is crucial for smooth reading.

Section 3.3 suffers from a repetitive opening. The first two sentences both state the goal of bridging the human-robot gap and transferring skills, effectively saying the same thing twice. Merging these would tighten the paragraph and improve the pacing. Similarly, in Section 4.3, the sentence "Using this representative subset ensures an unbiased evaluation..." contains a grammatical error where the subject ("subset") does not logically perform the action ("ensures").

This is a classic dangling modifier that disrupts the sentence's logic.

In Section 4.2, the introductory sentence to the task list is overly long and complex, burying the main point. A simpler lead-in would allow the reader to immediately engage with the list of tasks. Finally, in Section 3.4, the transition between the bolded sub-headers and the subsequent text is slightly abrupt; ensuring the text flows naturally from the header's noun phrase would improve readability.

These issues are minor and easily fixable with careful editing, but addressing them will ensure the reader can move through the technical details without unnecessary cognitive load.

Action items.

- [**writing**] The paper is generally well-structured and the narrative flow is strong, particularly in the Introduction where the problem and solution are clearly delineated. However, there are several instances where sentence construction, redundancy, or undefined variables create minor friction for the reader. In Section 3.1, the variable I_V appears in an equation without prior definition, while the surrounding text refers to I_{ref}. This forces the reader to pause and infer that I_V is the referenc